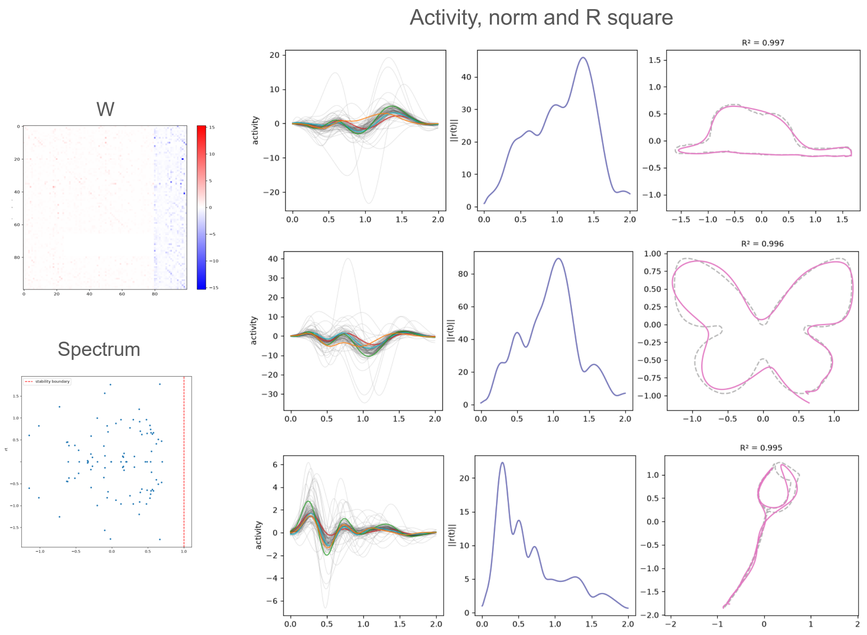
Robustness of Spectral and Dynamical Properties in Cortical Network Models Operating Near Instability

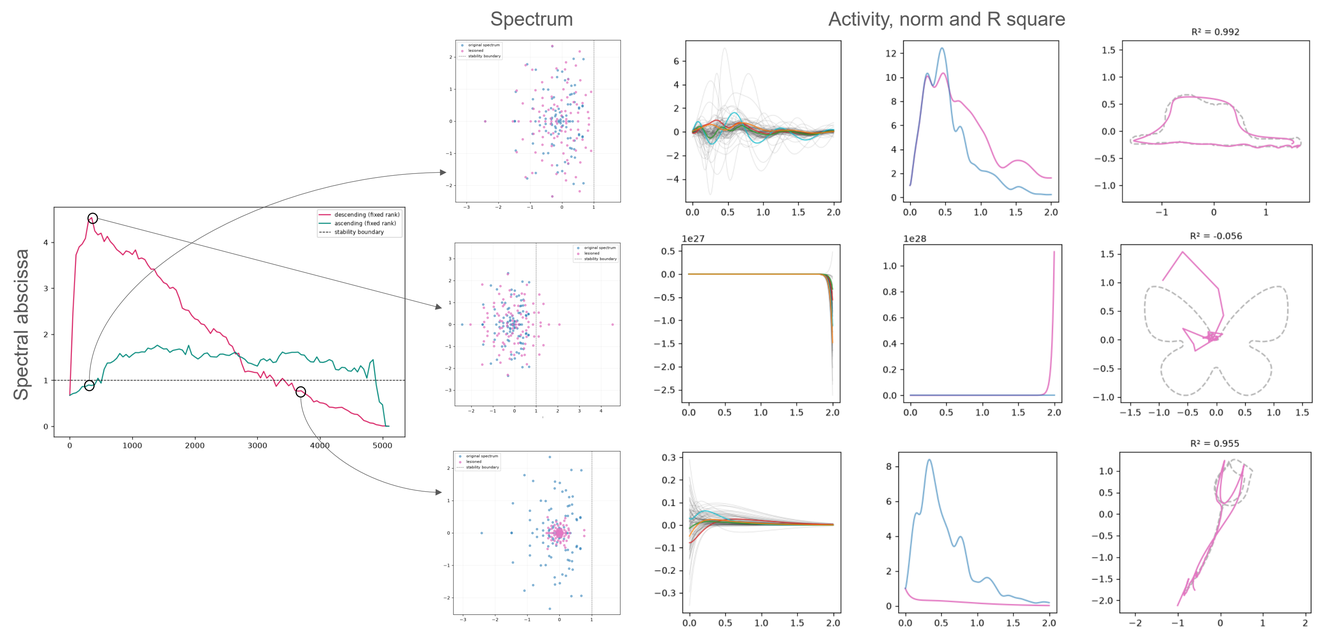
V. Scozzi, N. Condruz, T. P. Vogels

Non-lesioned Network



An example of the networks we considered along with its spectrum. On the right, the activity produced by 3 of its most amplifying initial conditions as well as the decoded shapes.

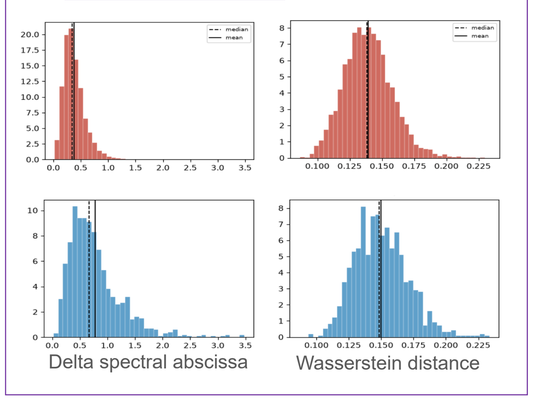
Lesioned Networks



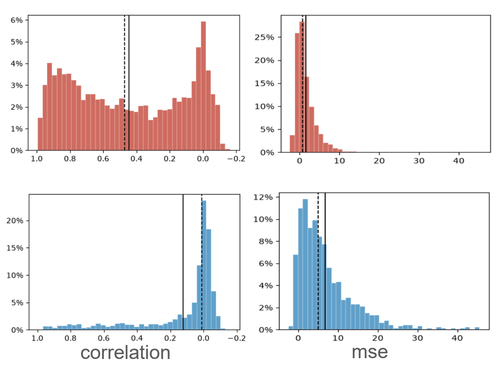
We progressively lesioned each network and considered how its spectrum and dynamics changed. Depending on the order of the removals, the networks could handle a number of lesions before becoming unstable. Once the abscissa crosses the threshold, the activity grows exponentially. As lesions accumulate and neurons become progressively more isolated, the spectrum collapses around 0 and the activity decays exponentially.

Neurons' Impact

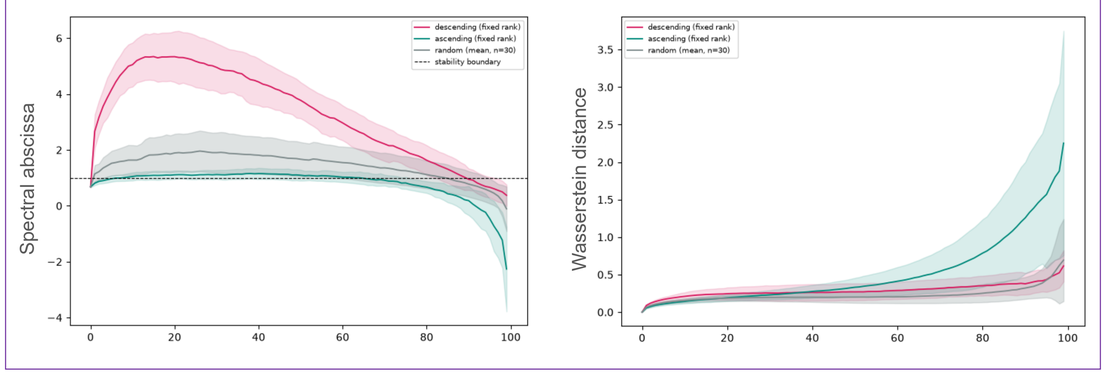
spectrum



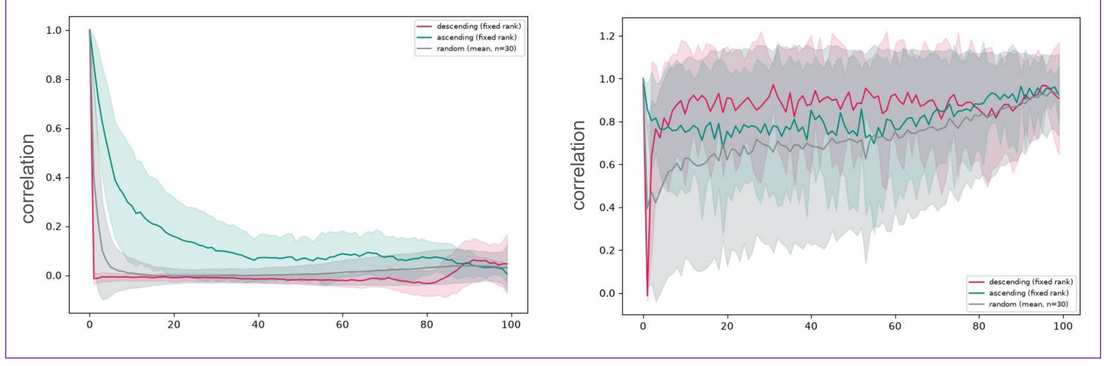
dynamics



Spectrum cumulative



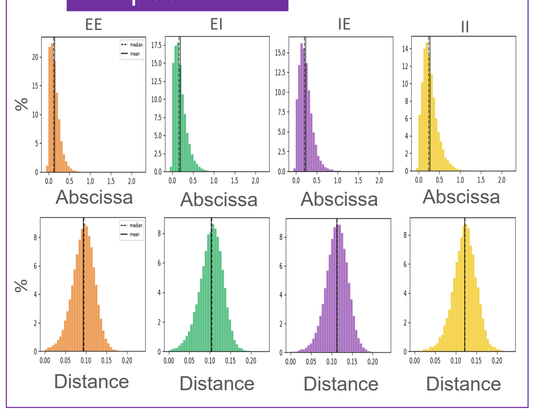
Dynamics cumulative



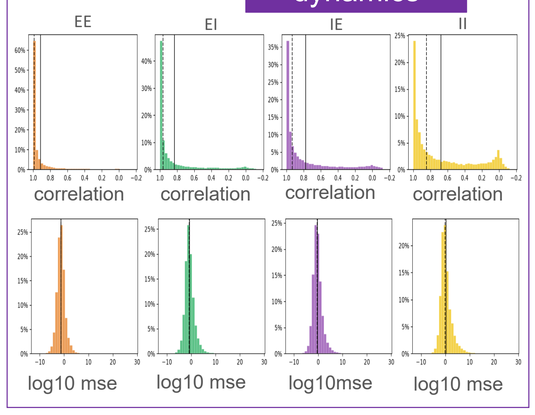
Inhibitory neurons matter more for stability, and consequently for dynamics, than excitatory neurons. The curves show the effects on spectrum and dynamics of progressive neurons' removal in 3 different orders.

Synapses' Impact

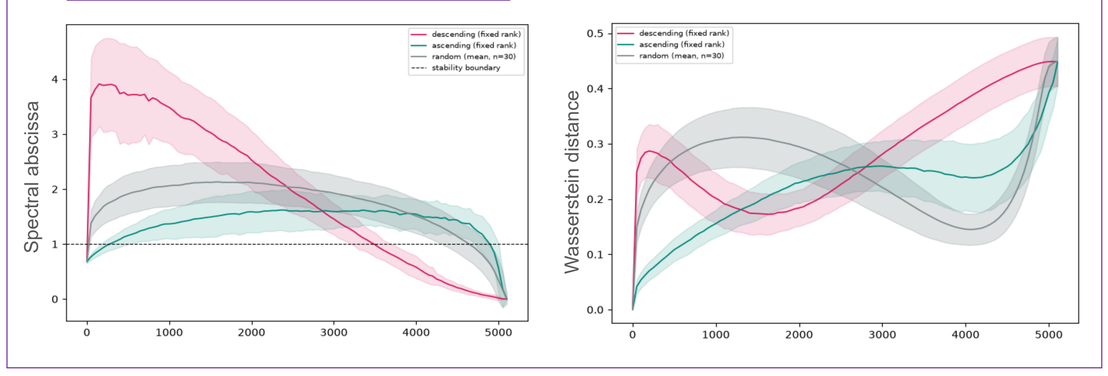
spectrum



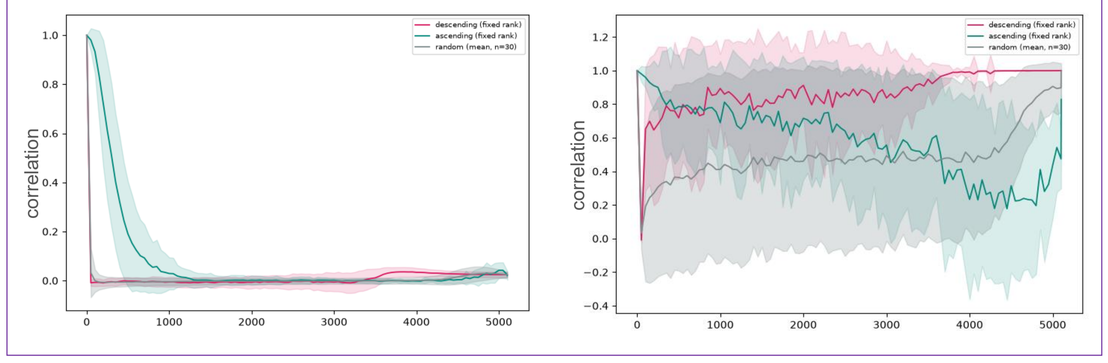
dynamics



Spectrum cumulative



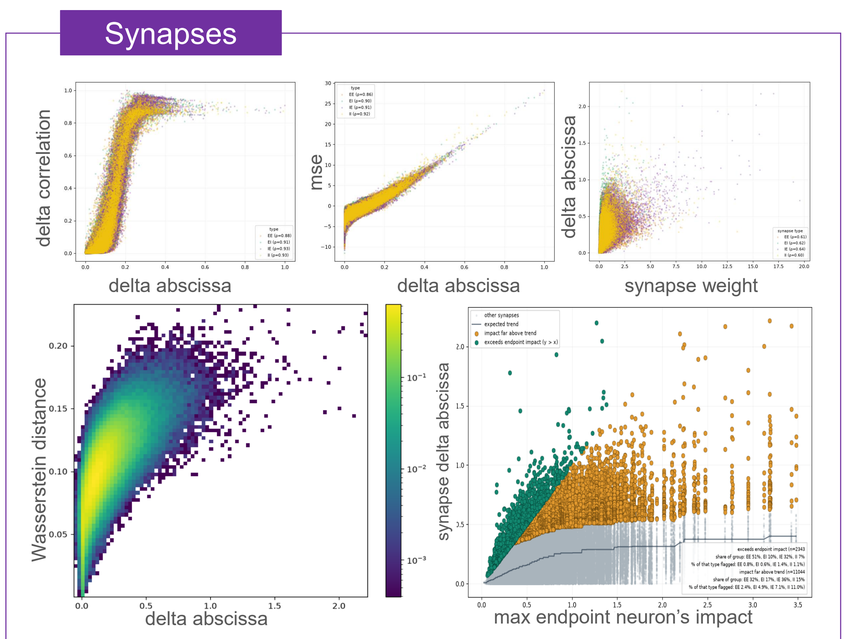
Dynamics cumulative



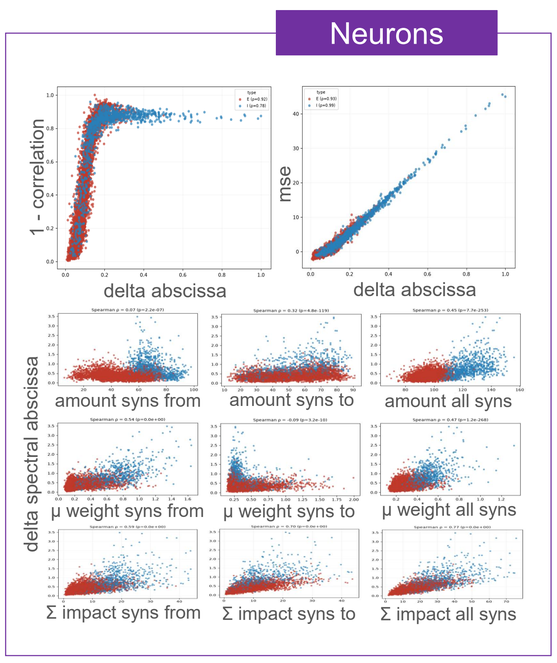
Synapses from inhibitory neurons have a larger impact than synapses from excitatory ones. II synapses have the largest impact. The curves show the effects on spectrum and dynamics of progressive synapses' removal in 3 orders.

Some further Analysis

Synapses



Neurons



References

[1] Hennequin G, Vogels TP, Gerstner W. (2012) Non-normal amplification in random balanced neuronal networks, 10.1103/PhysRevE.86.011909
[2] Bondanelli G, Ostojic S. (2020) Coding with transient trajectories in recurrent neural networks, 10.1371/journal.pcbi.1007655
[3] Christodoulou G, Vogels TP, Agnes EJ. (2022) Regimes and mechanisms of transient amplification in abstract and biological neural networks, 10.1371/journal.pcbi.1010365
[4] Hennequin G, Vogels TP, Gerstner W. (2014) Optimal Control of Transient Dynamics in Balanced Networks Supports Generation of Complex Movements, 10.1016/j.neuron.2014.04.045
[5] Murphy BK, Miller KD. (2009) Balanced amplification: a new mechanism of selective amplification of neural activity patterns, 10.1016/j.neuron.2009.02.005

Institute of Science and Technology Austria